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## EFFECT OF ASCORBIC ACID AND MENTHONE ON THE CASPASE 3 IN THE SPERM CELLS OF ACYCLOVIR TREATED RATS

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### ABSTRACT

**Background and aim:** Acyclovir (ACV) is a common drug used for treatment of Herpes virus infection. In this study, the role of this drug on apoptosis in rat spermatozoa was investigated along with the effectiveness of two herbal antioxidants of vitamin C (VC) and menthone in the prevention of ACV adverse effects was compared.

**Materials and methods:** Albino Wister adult male rats with mean weight of 200-250 g were divided randomly into six groups. Treatment groups consisted of acyclovir (ACV) at the dose of 15mg/kg/d, ACV (15mg/kg/d) and VC at the dose of 20mg/kg/d, ACV (15mg/kg/d) and menthone at the dose of 100µl/d, ACV (15mg/kg/d) and menthone at the dose of 250µl/d, ACV (15mg/kg/d) and menthone at the dose of 400µl/d and control group ended the trial period without any treatment. The animals were anesthetized and epididymal sperm cells from each rats were isolated in PBS to evaluate the expression of caspase3 enzyme by flow cytometry method.

**Results:** The present results demonstrated a very strong supportive, protective effect along with impressive anti-apoptotic role of VC. The expression level of caspase 3 in VC receiving group was close to normal levels. Menthone, another antioxidant compound in the high dose level showed to have a strong anti-apoptotic power, against changes induced by ACV.

**Key words:** challenges and threats; health transformation plan; health and prevention; self-care; qualitative study.

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### Introduction

Multicellular animals consist of several trillion cells, and a large number of them die daily to maintain tissue homeostasis. During the early development of testis and in the mature testes proliferation, differentiation and apoptosis of the testicular germ cells are normally performed. Apoptosis is necessary for the death of abnormal and excessive germ cells<sup>(1)</sup>. Some drugs or chemical Compounds or chronic infections cause disruptive apoptosis and spermatogenesis defect. The impact of various toxic compounds and high doses of drugs on the death of germ cells have been investigated<sup>(2)</sup>. Some studies, reported that acyclovir the viral DNA polymerase enzyme and has little impact on the host cell repli-

cation process and in other studies the side effect of this drug on the DNA synthesis in the germ cells is established<sup>(3,4)</sup>. 9 to 33% of the acyclovir transport in the blood is by transporter proteins. This compound has a wide tissue distribution and the half-life of this drug in the body depends on the age group<sup>(5)</sup>. Estimation of free radical generation and antioxidants defense during the past few years has become an important issue of research in animals for the protection of cells against oxidative damage<sup>(6)</sup> and toxicity of chemotherapeutic agents generating probably oxygen-reactive species (ROS) which causes oxidative stress<sup>(7)</sup>. Natural herbal drugs have been increasingly prescribed for the prevention and treatment of various diseases<sup>(8,9)</sup>. Peppermint (*Mentha piperita* L.) is from mint (*Lamiaceae*) fam-

ily and is considered as a medicinal and aromatic plant species. Peppermint essential oil constitute menthol, menthone, methylacetat, menthofuran and pulegone<sup>(10,11)</sup>. It is easily cultivated and has economic importance, due to its ability to produce and store essential oil, whose main constituent is menthol, used in oral hygiene products, pharmaceuticals, cosmetics and foods. Menthol also has high antifungal and antibacterial potentials, one of the most demanded substances by the scents and essences industry<sup>(12)</sup>. Mint, is widely known as one of the medicinal herbs with significant antioxidant activity. Their use has become expanded worldwide for treating various kinds of inflammatory diseases.

Menthone is one of the monoterpenes of peppermint oil and its antioxidant activity protect the DNA<sup>(13)</sup>. Because of its antioxidant potential menthone is selected for this compensatory treatment<sup>(14)</sup>. Vitamin C (L-ascorbic acid) or anionic form of it, ascorbate are essential components to humans and animal species. Ascorbic acid and ascorbate which naturally exist in the body depend on the buffering PH of body fluids and are consumed by the cells. VC is a cofactor of at least eight enzymatic reactions, such as synthesis of collagen, keratin, neurotransmitters and tyrosine catabolism<sup>(15)</sup> and is an essential compound for normal spermatogenesis (16). Previous researches reported that antioxidant can protect DNA and other important molecules against oxidation and damage, and can improve the quality of sperm and increase the success of fertility in men<sup>(17,18)</sup>. Therefore in this study, the role ACV as a gonado-toxic agent on apoptosis in rat spermatozoa was investigated along with the compensatory effect of VC in comparison with menthone.

## Materials and methods

### Animals

In this study 108 adult male albino Wistar rats (8 weeks old and weighing  $250 \pm 10$ g) were obtained from animal facility of Pasture Institute of Iran. All animals were treated in accordance to the Principles of Laboratory Animal Care. The animals were kept under specific conditions on a constant 12 hour light/dark cycle with a controlled temperature of 22 to 24 °C. They were kept under hygienic conditions in standard cages. Standard pellet food diet and water was provided ad libitum, while a 12h light/dark cycle was maintained. The animals were allowed to acclimatise for one week before the start of experiment.

This study was an experimental study conducted in accordance with the internationally accepted principles for laboratory animal's use and care research and with institutional Guidelines of Ethical Committee.

### Drug treatment

The drug ACV (MYLAN Company, France) was dissolved in distilled water before injection for this experiment. Animals were randomly divided into six groups of 18 rats (N=18): One group as control group, second group was administered intraperitoneally dose of 15 mg / kg / d acyclovir in ACV group. The other groups administered with supportive treatment as followed: ACV/VC20 mg/kg/d, ACV/menthone 100  $\mu$ l/d and ACV and menthone 250  $\mu$ l/d and ACV and menthone 400  $\mu$ l/d. (menthone purchased from MERCK catalog number: 8.41059.0025) for 48 days. The dosages of ACV were similar to those used in human therapy.

### Sampling and isolation of spermatozoa

The rats were anesthetized with administration of 10% ketamine solution intraperitoneally, 24 h after the last treatment. At the end of the experiment the animals were sacrificed. The testis in the control and experimental groups were immediately removed. The epididymal tissue was dissected from each rat. the rat's sperm were collected by cutting the tail of epididymis, into small pieces in 1 cc in phosphate buffered saline (PBS) with 0.05% tween (PBST) solution and incubated at 37 °C in an atmosphere of 5% CO<sub>2</sub> to allow the sperm to swim out of the epididymal tubules. The sperms washed in the PBST by centrifuge (400g for 5 min) into a pellet and the supernatant solution was aspirated. For each sample,  $1 \times 10^6$  -  $2 \times 10^6$  cells was used as a final concentration in 50-100  $\mu$ l of staining buffer.

### Detection of activated caspase 3

Levels of activated caspase 3 were detected in spermatozoa using fluorescent-labelled secondary antibody system. Primary antibody (PB) used was anti-caspase 3 (abcam, ab4051) and secondary antibody was Goat anti-rabbit (FITC, sigma: F1262). The sperm cells were re-suspended in 100  $\mu$ l of 4% paraformaldehyde solution at 4°C for 20 min. The cells were washed once with 1ml of PBST (400g/5min), permeabilization was carried out by adding 100  $\mu$ l (0.1-0.2%) Triton X -100, incubating for 10 min at room temperature. The cells washed once with 1 ml of PBST (400g/5min).

For blocking unspecific antibody protein binding add 100  $\mu$ l secondary host serum or BSA 1-2% for incubate 15min 37°C or 30 min at room temperature or 45 min 4°C. The cells were centrifuged and washed with 1 ml of PBST (400g/5min) twice 5 min. The sperm cells pellet are separated into samples tubes needed as Iso type control and test. The primary PB was added (dilute PBS/Tween-0/05%) into test tubes and incubated for 30-45 min in 37°C or (over night at 4°C). The sperm cells incubated for 5 min with 1 ml of PBST and centrifuged (400g/5min). The secondary Ab was added (dilute PBS-/Tween-0/05%) and incubated for 30 min in 37°C or 45 min 4°C. For isotype control samples just use secondary Ab. The cells are washed once with 1 ml of PBST (400g/5min) twice 5 min. For flowcytometry analysis, test samples was prepared with primary and secondary Ab, secondary Ab for isotype control samples and paraformaldehyde fixed samples without any antibody<sup>(19)</sup>.

### Statistical analysis

Statistical analysis of the data from flowcytometry was analyzed using the One-Way Analyses of Variance (ANOVA) method and LSD post hoc test using SPSS software version 16. Value of P less than 0.01 were considered as statistically significant.

### Results

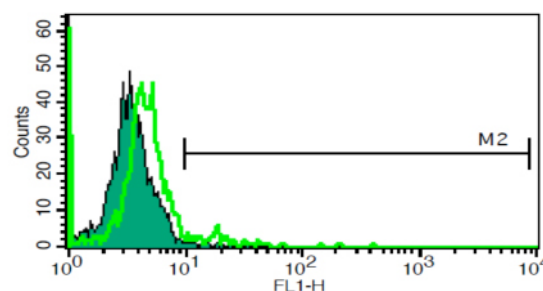
The obtained results are illustrated in the table 1 along with flowcytometry graphs of caspase 3 expression in the five test groups compared to the control group. The spermatozoa belonging to the group receiving acyclovir, caspase 3 expression increased up to 82% ( $p < 0.001$ ) and this indicates that apoptosis is activated in sperms cells (fig. 1).

| groups  | N  | Mean% | Std. Error |
|---------|----|-------|------------|
| ACV     | 18 | 82.11 | 0.478      |
| ACV/VC  | 18 | 5.44  | 0.202      |
| ACV.M1  | 18 | 55.44 | 0.755      |
| ACV.M2  | 18 | 18    | 0.42       |
| AVC.M3  | 18 | 12.06 | 0.318      |
| Control | 18 | 5     | 0.243      |

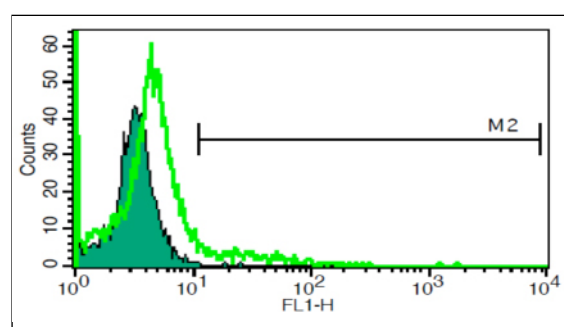
**Table 1:** comparison of the mean percentages of caspase 3 expression in the sperm cells in test and control groups. ( $p < 0.001$ ).

The results of the flow-cytometry analysis also showed that the two antioxidants used in this

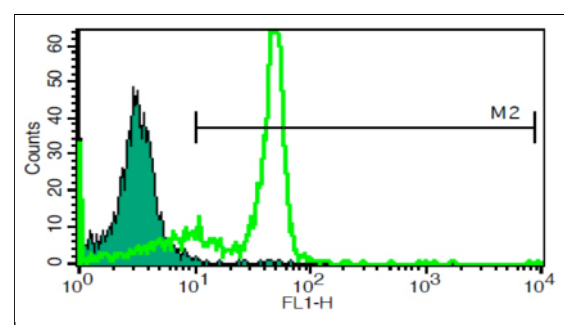
research has a significant protective effect. VC co-treatment in the acyclovir receiving groups reduced caspase 3 expression amount to 5.44% ( $p < 0.001$ ) that was very close to the control group (fig. 2,3).



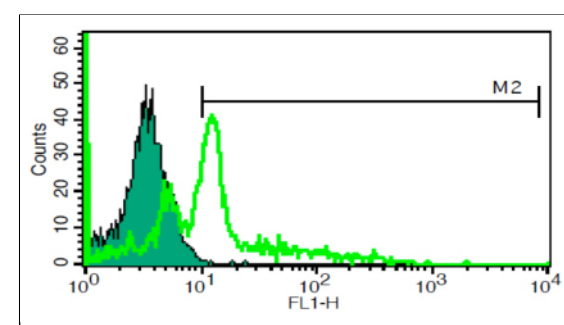
**Fig. 1:** Control 5%.



**Fig. 2:** Vitamin C 7%.



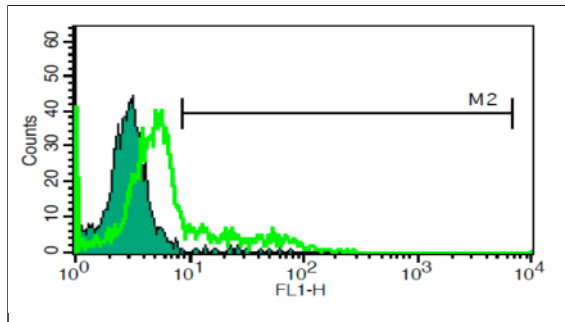
**Fig. 3:** Acyclovir 83%.



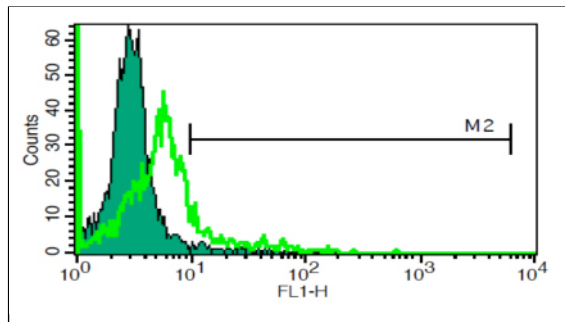
**Fig. 4:** Menthone 55%.

In the three groups with menthone co-treatment a dose-dependent decrease of caspase 3 was observed. Menthone at a dose level of 100  $\mu$ l /d reduced the caspase 3 expression to 55.44%

( $p < 0.001$ ) (fig. 4) where as in the dose level of  $250 \mu\text{l/d}$  of menthone caspase 3 expression reduced to 18% ( $p < 0.001$ ) (fig. 5). The relative decrease in the caspase 3 expression at the highest dose of menthone was 12.06% ( $p < 0.001$ ) which reflects the good potential ability of it to support the cells against apoptosis (fig. 6).



**Fig. 5:** Menthone 2 18%.



**Fig. 6:** Menthone 3 12%.

**Figure 1-6:** Comparison of the mean percentages of caspase 3 expression. In each of the flow cytometry graphs the curve of which has a dark green color Represents the caspase 3 expression level of the negative control sample and bright curve indicates the level of caspase 3 expression in the assessment samples. Charts show significant differences at  $p < 0.001$ .

## Discussion

In the present study, the antioxidant effect of ascorbic acid and menthone was compared along with the adverse effects of acyclovir on reproductive system in male rats. ACV is an antiviral drugs used by men of different ages, causing cytotoxic and genotoxic risks to germ cells. This side effects can be transferred to next generations, inherited in the offspring in the form of different disease. The role of oxidants and antioxidants have attracted widespread interest in nutrition research, medicine and biology. It has been reported that regular generation of pro-oxidants, with oxygen free radicals, can be an essential attribute of aerobic life<sup>(20)</sup>. An imbal-

ance in the pro-oxidant and oxidant has been referred as oxidative stress. Reactive oxygen species (ROS) are considered as free radicals because of presence of one unpaired electron like superoxide ion, nitrogen and hydroxyl radical. They are naturally present in organism, and are confined to cell compartments but counterbalanced by natural antioxidant acting as free radical scavengers<sup>(21,22)</sup>. On the other hand ACV is an important inducer for reactive oxygen species (ROS) production, ROS causes damage to the spermatozoa DNA, resulting in increased apoptosis of these cells. The production of ROS is greatly dependent to the various environmental and life style factors such as pollution, smoking and drug consumption. Various endogenous antioxidants belonging to both enzymatic and non-enzymatic groups can remove the excess ROS and prevent oxidative stress<sup>(23)</sup>.

Caspases are proteases enzymes containing the amino acid cysteine in their active sites<sup>(24)</sup>, Activated caspases mediate degradations of cell quality by cleaving to their intercellular protein targets<sup>(25)</sup>. Caspases involved in apoptosis act either as initiators (caspases 8, 9, 10) or as effectors (caspases 3, 6, and 7) of apoptosis<sup>(26)</sup>. The most important caspase is caspase 3, it executes the final disassembly of the cell by cleaving a variety of cellular proteins and generating DNA strand breaks<sup>(27)</sup>. ACV in male rats is responsible for increase in the serum malondialdehyde (MDA) level, that is an important indicator of lipid peroxidation (LPO), leading to reproductive disorders caused by ROS in spermatogenic cells<sup>(28)</sup>.

LPO formed by ROS can damage cellular structure, motility, survival and metabolic functions of sperm and induces its apoptosis<sup>(29)</sup>. In our study caspase 3 expression was measured as an important index of apoptosis. This enzyme was observed to increase significantly in ACV treated group and our findings were concordant with previous studies. In the present study ,spermatozoa of the rat specimens belonging to the group receiving acyclovir, caspase 3 expression was found to increased up to 82% ( $p < 0.001$ ) ,indicates that apoptosis is activated in sperms cells. This result are supported by the findings of Weng et al (2002), who reported that the percentage of caspase 3 expression in the cells is an important indicator for distinguishing of apoptotic cells from therapeutic gain<sup>(30)</sup>.

In another study, reported that apoptosis inducer agent ACV, is one of the antiviral drugs that can cause decrease sperm count because of its cytotoxic-

city<sup>(31)</sup>. Acyclovir can initiate germ cell death in the seminiferous epithelium and diminished the sperm count and sperm motility<sup>(32)</sup>. In a previous study, acyclovir treatment of mice for more than 35 days duration causes spermatogenesis toxicity. Narayana et al indicated that ACV can remarkably damage the tail of sperms and inhibit their motility because abnormal sperms production is due to the acyclovir genotoxicity and mutations in the spermatogonia<sup>(33)</sup>. Acyclovir can defects the DNA synthesis during mitotic stage of spermatogenesis<sup>(34)</sup>. For decreasing these side effects we established four co-treatment groups consisting ACV/V C, ACV/menthone 100 $\mu$ l/d, ACV/menthone 200 $\mu$ l/d, ACV/menthone 400 $\mu$ l/d for comparing the anti-apoptotic potential of these two anti-oxidizing agents.

It has long been argued that ascorbic acid and plant extracts have the antioxidant and anti apoptotic effects, so we estimate the anti apoptotic potential of vitamin C along with menthone comparatively. Fraga et al, suggested that dietary ascorbic acid can effectively protects human sperm from endogenous oxidative DNA damage, that could affect sperm quality and increase risk of genetic defects<sup>(35)</sup>. In another study, they demonstrated that epididymal sperm concentration and plasma testosterone level significantly increased in the ascorbic acid treated rats and suggest that ascorbic acid supplementation improves reproductive traits of male rats that are associated with high fertility<sup>(36)</sup>. Ascorbic acid and ascorbate are powerful reducing agents for removal of ROS<sup>(37)</sup>.

It is indicated that receiving 90 mg of VC in daily diet is necessary and it is diagnosed that its deficiency causes a disruption of normal spermatogenesis cycle and decline in testosterone levels<sup>(29,38)</sup>. Extensive degeneration of spermatocytes have been observed in vitamin C deficient rats and treatment of rats with an antioxidant such as VC can compensate the tissue damage of oxidative stresses<sup>(39)</sup>. V C has a significant role against oxidative stress induced apoptosis in testicular spermatogonial cells<sup>(40)</sup>. Vitamin C is an important antioxidant in human plasma, capable of scavenging oxygen-derived free radicals and sparing other endogenous antioxidants from consumption<sup>(41-43)</sup>. In a study, the protective effect of ascorbic acid in patients treated with cyclophosphamide in diminishing some of its adverse effects was identified<sup>(44)</sup>. Therefore, as an offset against the side effects of acyclovir treatment, the VC was used in our study and obtained data suggesting that VC co-treatment in the acy-

clovir receiving groups reduced caspase 3 expression in the sperm cells amount to 5.44% ( $p < 0.001$ ) which was very close to the control group.

The result of the present study are supported by Vissers et al. In their experiments, they concluded that ascorbic acid can protect animal cells against the cytotoxicity induced by oxidative stresses, in reducing apoptosis, and diminish the expression of caspase-3. Their results also suggest a role for vitamin C in the regulation of apoptosis in an oxidant-sensitive step in the initiation phase of this process<sup>(45)</sup>. Rewei et al in 2002 indicated that some radical scavengers such as N-acetylcysteine, VC, vitamin E and folate causes reduction of apoptosis<sup>(46)</sup>. VC is powerful antioxidant naturally present in contact with sperm and its effect improve sperm quality and reduce sperm DNA fragmentation parameters and protecting DNA against strand breaks<sup>(46)</sup>. Thus, the consistent result for VC in our study in relation to ACV make the favorable association of VC with reduction of caspase 3 expression in the sperm more credible.

Although there are many studies on the anti-apoptotic impact of VC, but a menthone supplementation trail has not yet been conducted on the caspase 3 in the sperm cells of male rats. And we in this paper for the first time reported the anti-apoptotic effect of invivo administration of menthone and compared it with the anti-apoptotic impact of VC.

Essential oils are isolated from plants cells, presenting some biological activity including antimicrobial, anti-inflammatory activity and eliminating pathogenic microorganisms. Essential oils produced by plants have been traditionally used for respiratory tract infections, or even treatment of viral, bacterial and parasitic infections. One of these biological activity compounds presents in peppermint oil is menthone that has antibacterial and anti-inflammatory effects<sup>(47)</sup>.

The main compound of the plant responsible for its pharmacological effects are menthone, isomenthone, menthol, 1, 8-cineole, borneol, and piperitenone<sup>(48)</sup>.

Menthone is a volatile oils from mints and pennyroyal and occurs in small quantities in other essential oils, and its safety in-use as a food ingredient is indicated<sup>(49)</sup>. In an invitro study, menthone increases cell proliferation and decreased IL-13 levels in leukocytes<sup>(50)</sup>. Monoterpenes such as menthone have a cytotoxic effect against the human tumoral cell lines<sup>(51)</sup>.

Monoterpenes have chemo preventive properties and have an antioxidant, antimutagenic and antiproliferative activity. They can enhance enzyme induction and detoxification<sup>(52, 53)</sup>.

In this study, pure L- menthone administered in three different doses as a compensatory co-treatment with ACV on the caspase 3 in the sperm cells of male rats. The flowcytometry observations indicated that this compound can reduce the percentage of caspase 3 expression in the rat spermatozoa in a dose dependent manner.

In conclusion, the reliability and validity of the assessment of the anti-apoptotic effect of invivo administration of highest dose of menthone was observed to reflect good potential ability, in making relative decrease in the caspase 3 expression to support the sperm cells against apoptosis.

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